

Hayden Gillen

Professor Johnson

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Project 2.4 A-Level Reflection

I originally ran the simulation and received an output of “3500 visit and 1780 left early”. With the help of comments in the code and the interlude in the instructions I was able to understand how the simulation works. Riders enter the park and choose a ride and if there are less than 150 people in the line they will join the queue. The ride goes and takes `RIDE_TIME` to run then adds more riders to the ride from the queue. Each ride can only hold `CAPACITY` riders before it goes off and the ride only waits for four minutes after the first rider enters before it takes off.

In `Start()` in the `CarnivalSimulation` class I added code to collect the amount of people waiting on the ride as a percentage of total ride capacity. The rides do not appear to be running at capacity as the average ride capacity is usually between 11% and 43%. The rides with a smaller capacity, on average, have a higher percentage of riders. This makes sense because it is easier to fill 30 seats than 120 seats since less people are needed create a larger percentage. Each rides maximum value was 100% capacity so at least one time during the day the ride ran at full capacity. Similarly, each rides minimum was 0% which means during the day at some point there was nobody on the ride. Additionally, there was a strong correlation between the average percentage of ride capacity and the total number of rides that the attraction gave. Rides with lower capacities are easier to fill and therefore are likely to complete more cycles than rides that take 120 out of 150 possible people from the queue. If there are less than 120 people in the queue

the ride will have to wait the full 4 minutes before dispatching, but those rides that only require 30 are likely to fill before the 4 minutes and be sent off earlier.

To improve ridership we could increase NUM_RIDES, while ride times do get longer, there are more of them to hold people and there has to be some optimal number of rides in park to prevent the most people from leaving early. Additionally, rides with 120 capacity run, on average, at 11% capacity about 277,000 times per day which means it is moving about 3.9 million riders per day. Rides that carry 30 run, on average, at 43% capacity about 387,000 times per day which means it can move about 4.4 million riders per day. In between at 60 capacity it carries, on average, 26% capacity for about 360,000 rides per day which means it can move about 5.7 million riders per day. The relationship between ride capacity and ridership is parabolic, which means that there is some absolute maximum that can be determined which is moves the most riders. If every ride had the capacity of this number than each ride could move the most riders per day thus increasing ridership. More simply, the queues could be extended to hold more people to prevent people from leaving the carnival.